

TASTE SENSORS' WORKING AND APPLICATIONS

Taste sensors, also known as electronic tongues, are particularly useful in food, beverage and pharmaceuticals industries for qualitative and quantitative analysis



Astree electronic tongue from Alpha MOS based on ChemFET sensor technology

■ **V.R. ADKOLI**

A sensor is a device that changes various types of physicochemical properties of a target to a different state (usually electrical signals). Our body parts are also sensors. For example, we have sensors for sight, hearing, touch, taste and smell.

Taste sensors, also known as artificial tongues or electronic tongues, measure and compare taste. These provide a quantified, objective scale

and find applications in food, beverage and pharmaceuticals industries.

Human tongue has about 10,000 taste buds with five taste sensations: sweet, bitter and savory (also called umami), which work with a signal through G-protein coupled receptors, and salty and sour, which work with ion channels.

Electronic tongues are analytical instruments. Some electronic tongues work using multiple least-square technique. Some scientists have also looked at using acoustic devices, with

so-called quartz crystal microbalances—said to be one of the first sensors that work by measuring frequency changes due to mass changing on device surface. Shear horizontal surface acoustic wave (SH-SAW) micro sensors are used to analyse complicated metallic tastes.

Principle of working

Liquid samples are directly analysed without any preparation, whereas solids require preliminary dissolution before measurement.

Reference electrode and sensors are dipped for 120 seconds in a beaker containing a test solution. A potentiometric difference between each sensor and a reference electrode is measured by means of a measuring device and recorded by an e-tongue software.

An electronic tongue module for the food and beverage industry may have a sensor board (having 0 to 28 channels), acquisition board (acquisition frequency of 50 Hz), power supply card of 220V-240V AC, and software for soft control and monitoring the sensor array system.

Toko & Hayashi Lab is the first in the world to successfully develop wafer-like taste sensors.

Twelve sets of MOS array with pattern recognition software have been used for discriminating between Brazilian and Columbian coffee.

Oxide gas sensors are used for alcohol and tobacco testing.

Sour taste sensors have been